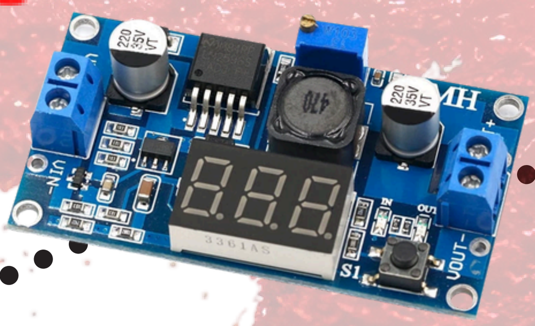
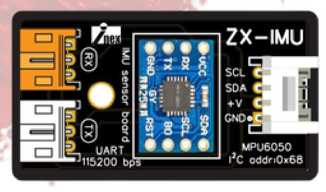
RoboCup Rescue

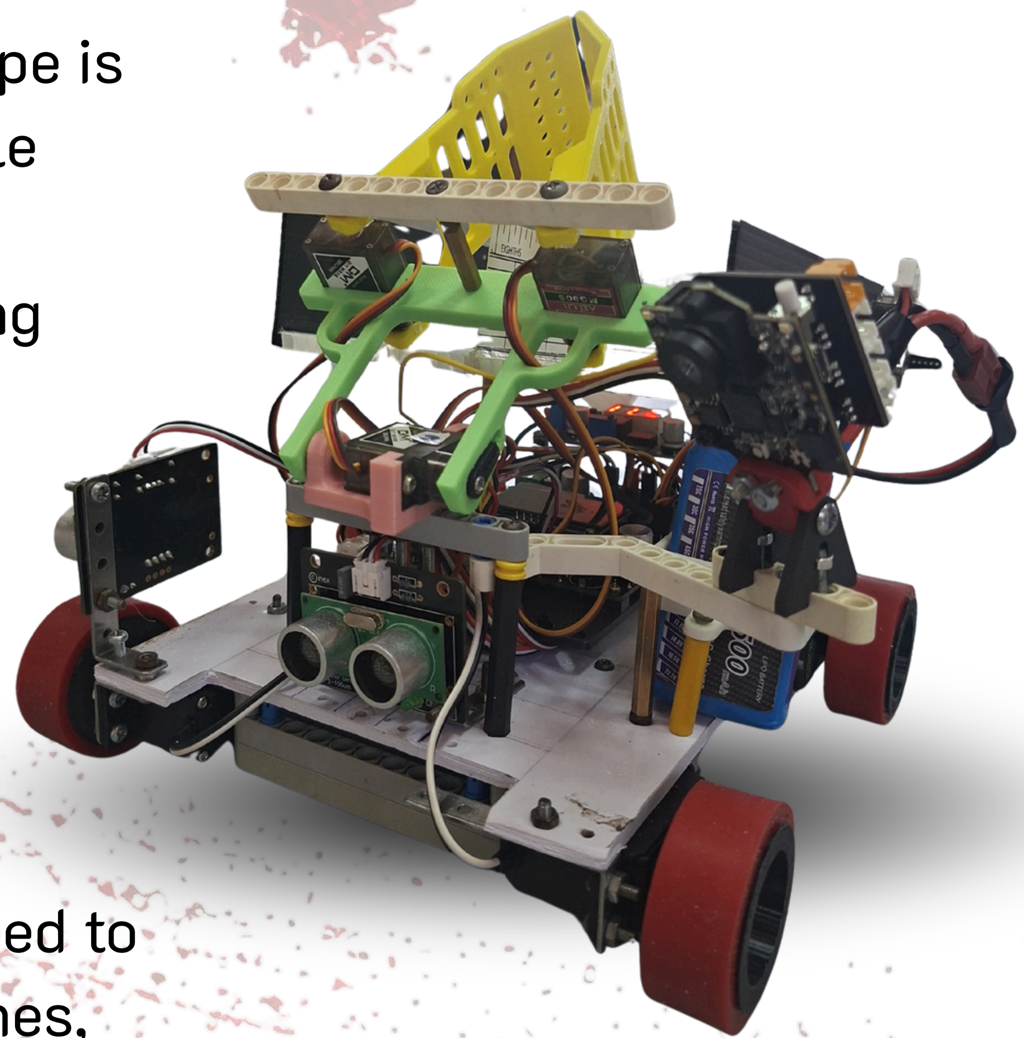
CNT-ROBOT
From Thailand

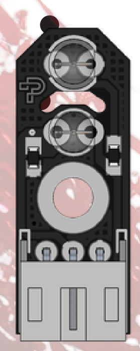


Lm2596 DC-DC
converter is used to regulate and stabilize the voltage before supplying power to the robot

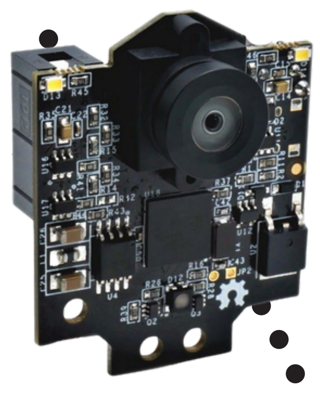


Gyro ZX-IMU
The ZX-IMU gyroscope is used for precise angle measurement and accurate robot turning



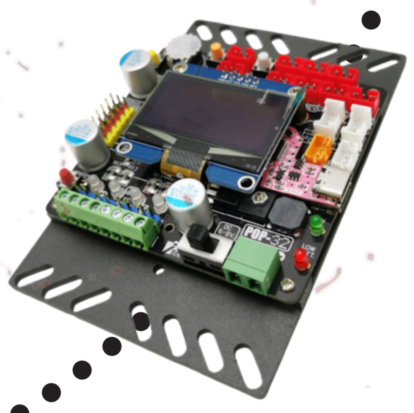


SENSOR SEN002
line-tracking sensor is used to detect black and white lines, enabling the robot to follow the designated path



Pixy 2.1 CMUcam5
camera used for ball detection and color classification

ZX-Sonar1M
for obstacle and wall detection



POP-32i
to process and execute programmed commands

MG90S servo
Used to control the gripper and release balls from the storage box

Motor PG-300
to drive the robot according to received commands

Soft Ware

The robot is programmed in C/C++ on the POP32 platform. It uses a State Machine for task control, PID for line following, real-time sensor processing for navigation, obstacle avoidance, intersection detection, gap recovery, and MG90S servo control for rescue ball delivery.